# Equality of Opportunity

## Economics of Public and Social Issues

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# Outline

1. Geographical Variation in Upward Mobility in America
2. Causal Effects of Places vs. Sorting

Lecture based primarily on the following paper:

*Chetty, Friedman, Hendren, Jones, Porter. "The Opportunity Atlas: Mapping the Childhood Roots of Social Mobility" NBER WP, 2018*

# Quiz: AI and learning

# Today's quiz

Three questions on Strömberg, Lei and Wu (2026), *The Generative AI Learning Penalty*.

- Answer in **iClicker** — scan the QR code on screen or open the app
- One answer per question; we look at the results together before moving on
- This is a **graded quiz**: you are scored on **getting it right**
- Quizzes are **12%** of your grade, and your **two lowest scores are dropped**

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Two of the three ask *how the study was done*, not just what it found — that is the part that will matter for your own projects.

# Q1. Where do the data come from?

**A.** A survey in which students remembered their past grades

**B.** Administrative records from the county education bureau — the same digital systems that log exam scores, homework scores, and homework timestamps

**C.** An experiment in which researchers assigned some students to use AI

**D.** Test scores published online by the Chinese government

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**Answer: B.** 26,811 students — 90% of the county. The homework platform records *when* each assignment was opened and submitted, which is how the authors can spot homework that was finished very fast and scored very well.

# Q2. Nobody was assigned to use AI. So how can the authors blame AI?

**A.** They compare AI users with non-users at a single point in time

**B.** They compare each student's scores **before vs. after she started using AI**, then subtract whatever happened over those same months to students who **never** started

**C.** They asked students whether they felt AI had hurt their grades

**D.** They compare schools that banned AI with schools that allowed it

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**D.** They compare schools that banned AI with schools that allowed it

**Answer: B.** Why subtract the never-users? Exams get harder, seasons change, students mature — the never-users soak all of that up. And the two groups moved **in parallel before** anyone adopted, which is what makes the comparison believable.

# Q3. What happened after students started using AI?

**A.** Homework scores and exam scores both went **up**

**B.** Homework scores and exam scores both went **down**

**C.** Homework scores went **up** and homework took **less time**, but exam scores went **down**

**D.** Homework was unchanged, but exam scores went **up**

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**Answer: C.** Homework scores **+18%**, time per assignment **64 → 45 minutes**, closed-book exam scores **−20%**. The work got easier. The learning did not happen.

# So what?

- The students who lost the most were the ones who used AI to **finish faster**
- The students who used AI **and kept their time on the task** scored about the same as everyone else
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# So what?

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Keep this in mind on Project 0: the goal is not a finished file, it is being able to **explain** the file.

# Geographical Variation in Upward Mobility

# Differences in Opportunity Across Local Areas

- How do children's chances of moving up vary across areas in America?
  - Are there some areas where kids do better than others? If so, what lessons can we learn from them?
- Recent studies have used big data to measure how upward mobility varies based on where children grow up

# The Opportunity Atlas: Data Sources and Sample Definitions

- Data sources: Anonymized Census data (2000, 2010, ACS) covering U.S. population linked to federal income tax returns from 1989-2015
- Link children to parents based on dependent claiming on tax returns
- Target sample: Children in 1978-83 birth cohorts who were born in the U.S. or are authorized immigrants who came to the U.S. in childhood
- Analysis sample: 20.5 million children, 96% coverage rate of target sample

# Measuring Parents' and Children's Incomes in Tax Data

- Parents' household incomes: average income reported on Form 1040 tax return from 1994-2000
- Children's incomes measured from tax returns in 2014-15 (ages 31-37)
- Focus on percentile ranks in national distribution:
- Rank children relative to others born in the same year and parents relative to other parents

# Intergenerational Income Mobility for Children Raised in Chicago

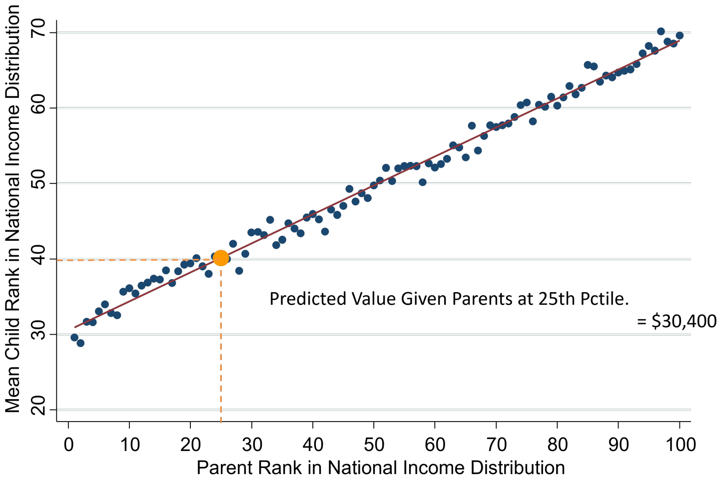


Figure 1: Source: Chetty, Hendren, Kline, Saez (QJE 2014)

# Intergen. Mobility for Children Raised in Hypothetical Census Tract

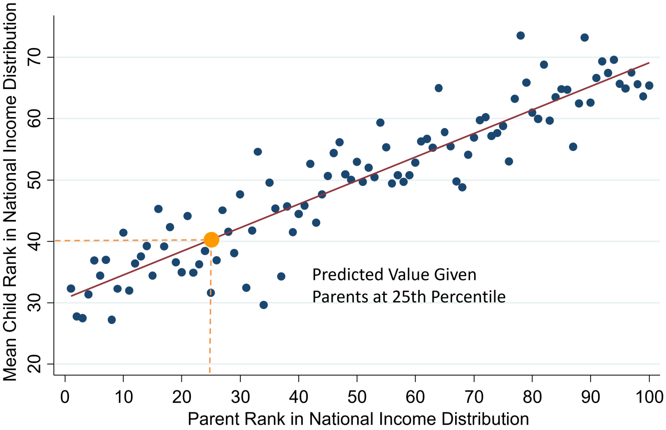


Figure 2: Source: Chetty, Hendren, Kline, Saez (QJE 2014)

# Estimating Children's Average Outcomes by Census Tract

- Run a separate regression using data for children who grow up in each Census tract in America
- In practice, many children move across areas in childhood
- Weight children by fraction of childhood (up to age 23) spent in a given area

# The Geography of Upward Mobility in the United States

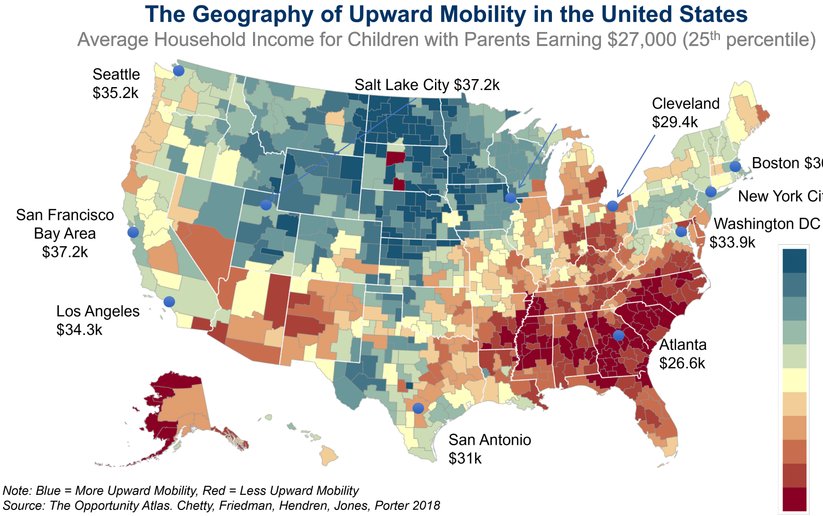


Figure 3: Source: Chetty, Friedman, Hendren, Jones and Porter (2018)

# Poll: what about San Antonio?

You have just seen the map for the whole country. Before we open the Atlas:

> ***For a child growing up in a low-income family here in San Antonio, how do the chances of moving up the income ladder compare with the rest of the country?***

**A.** Higher than the national average

**B.** Lower than the national average

**C.** About the same

This one is a **poll, not a quiz** — you get credit for answering, and nothing is taken off for guessing wrong. I want to see what the room thinks *before* we look at the data.

# Jump to Opportunity Atlas website

- https://www.opportunityatlas.org/
- Explore Durham and San Antonio briefly